



—

INFOSYS 700

Internet of Things

not connected with each other

wearable tech: watch, glass, earphone, ^{smart,} ring,

Solar, 停電. 断行↑

IOT

THING + IT $\begin{matrix} \text{Hardware} \\ \text{Software} \end{matrix}$ = Thing-based $\begin{matrix} \text{Physical} \\ \text{Local} \end{matrix}$ function + IT-based $\begin{matrix} \text{Digital} \\ \text{Global} \end{matrix}$ service

 Bulb IoT stack → Light Security

 Bin IoT stack → Storage capacity Automatic replenishment

 Towing vehicle Predictive maintenance, optimization

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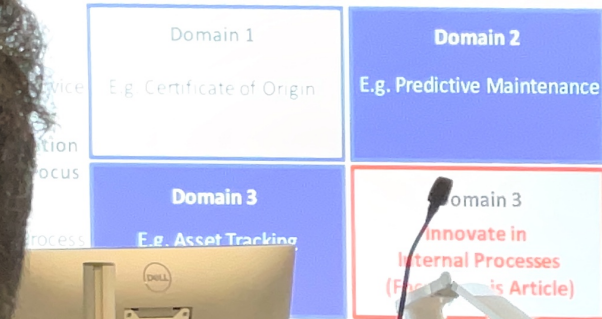
 Tractor IoT stack → Towing vehicle Predictive maintenance, optimization

IoT cloud

process

Security
efficiency

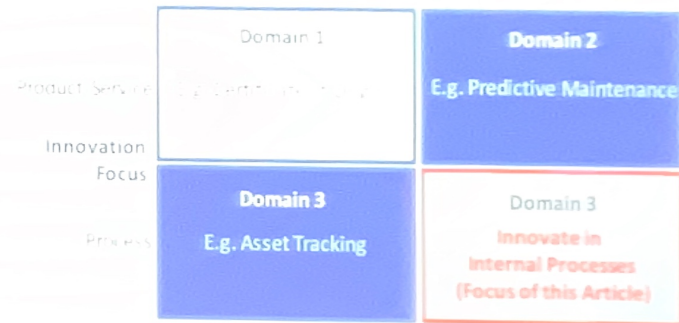
Four Domains Leveraging IoT Digital Data Stream



Source of Digital Data Streams

Bäcker, D., Flesch, F., & Wortmann, F. (2019). Driving process innovation with IoT field data. *MIS Quarterly Executive*, 18(3), 191–207. <https://doi.org/10.17705/2msex.00016>

Four Domains Leveraging IoT Digital Data Stream



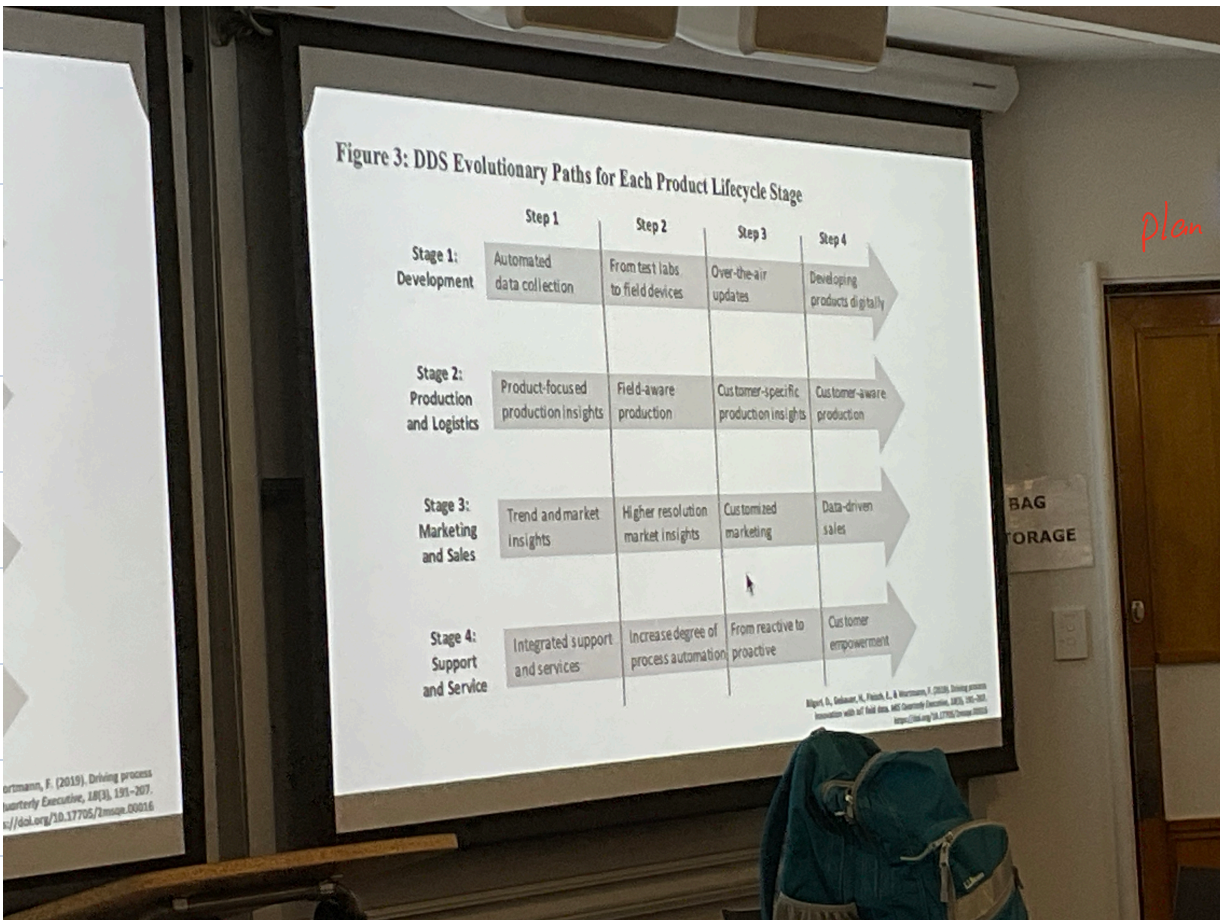
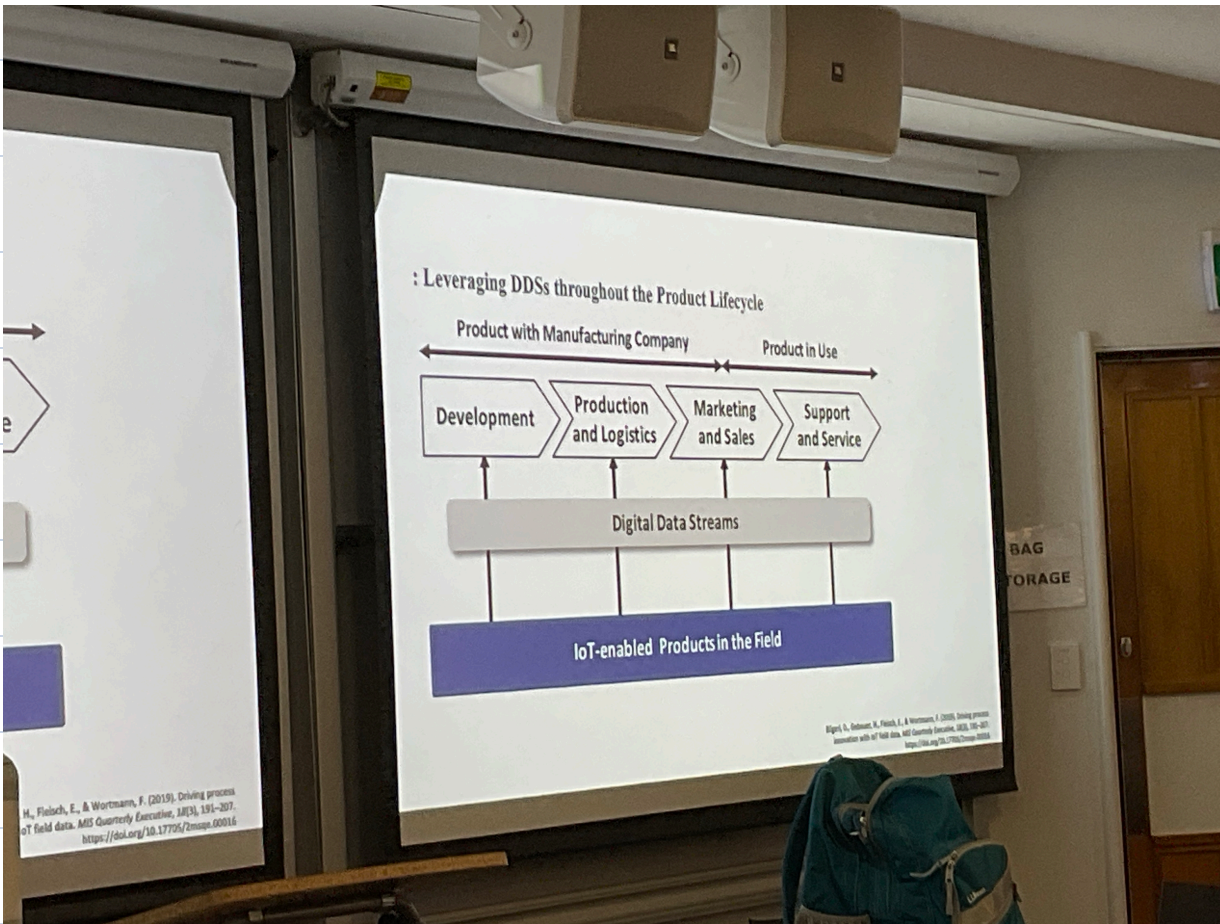
Supply-chain Data

Source of Digital Data Streams

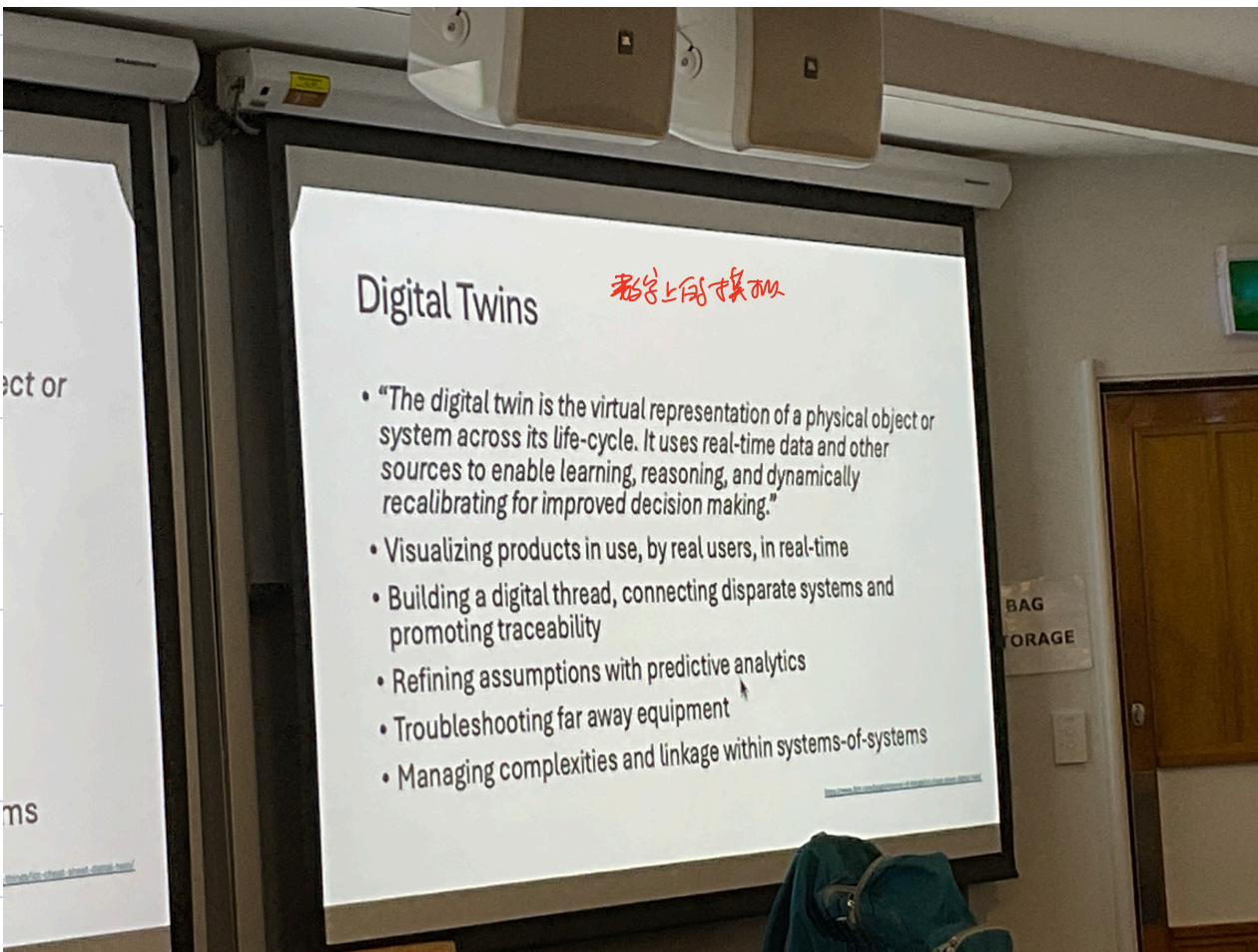
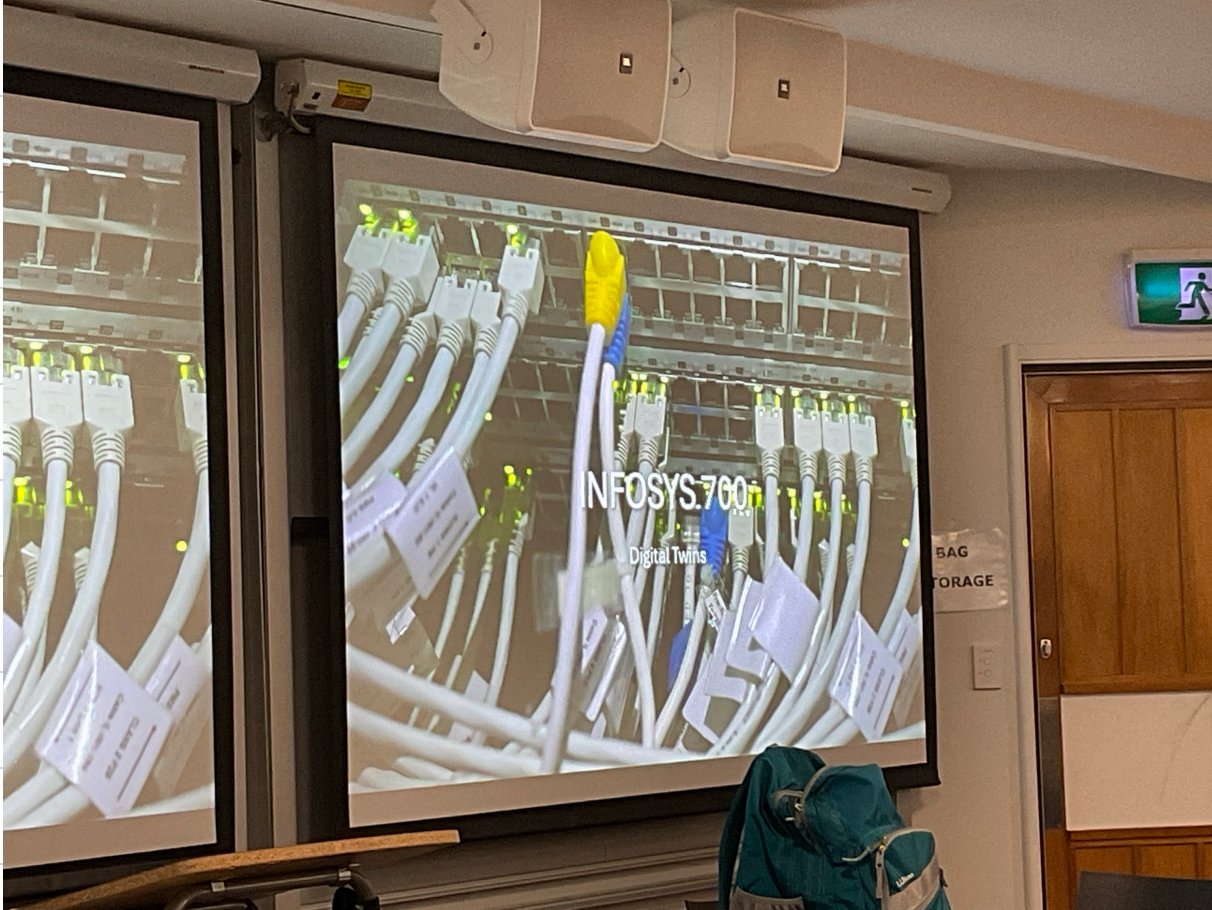
Field Data

Bäcker, D., Flesch, F., & Wortmann, F. (2019). Driving process innovation with IoT field data. *MIS Quarterly Executive*, 18(3), 191–207. <https://doi.org/10.17705/2msex.00016>

poling



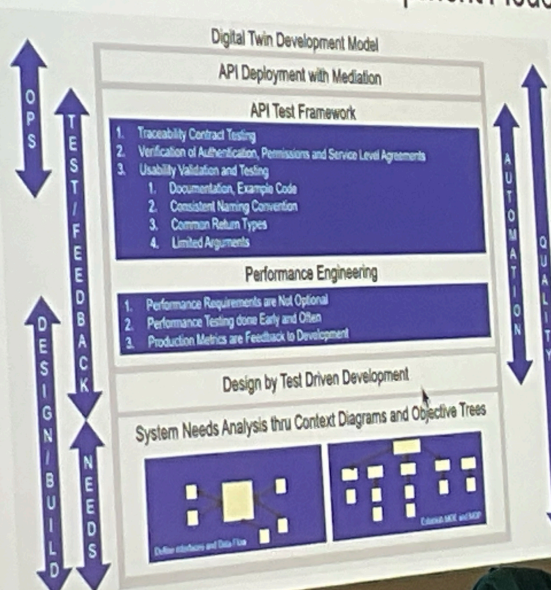
plan 13

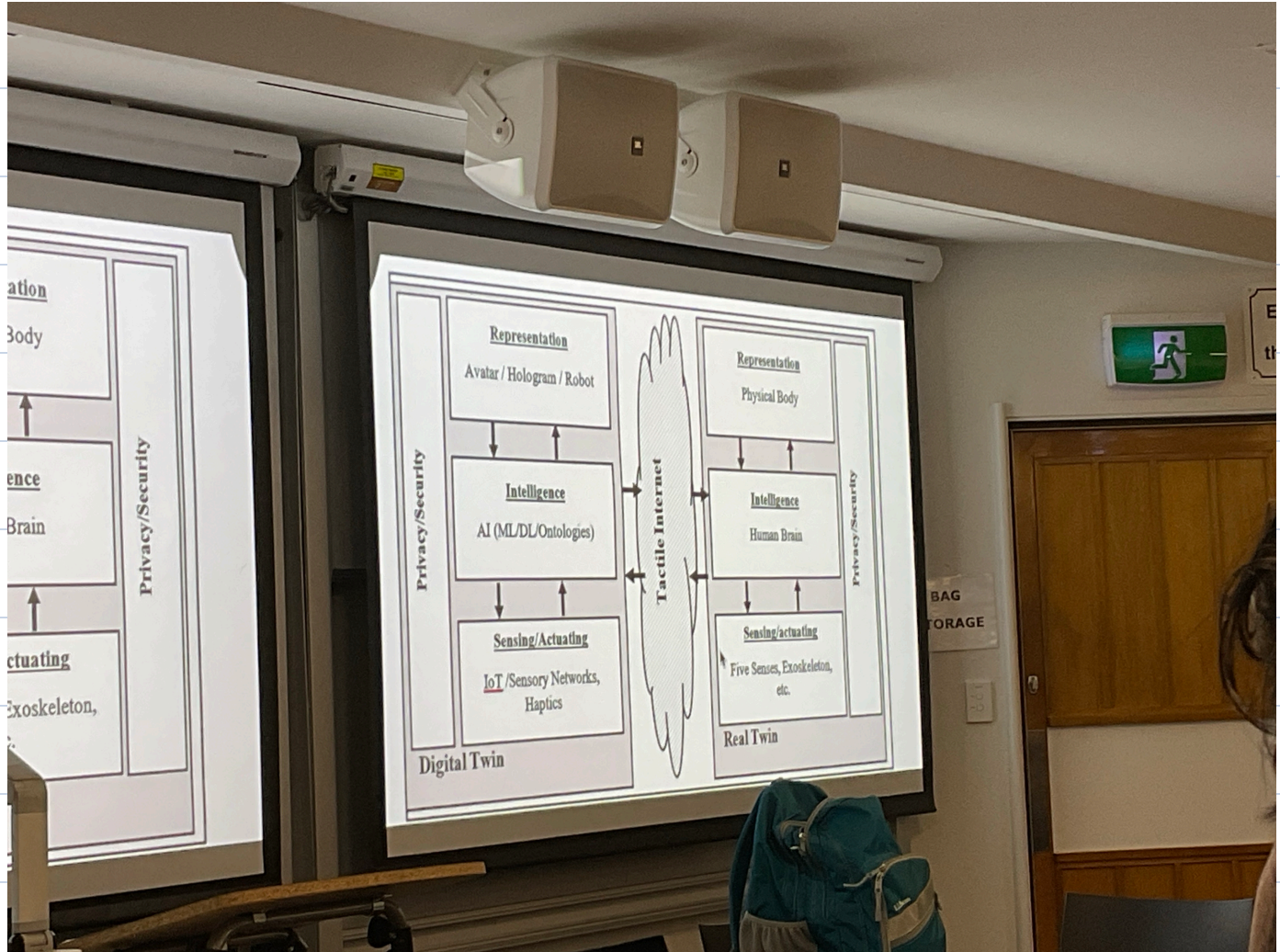


In terms of Information Systems

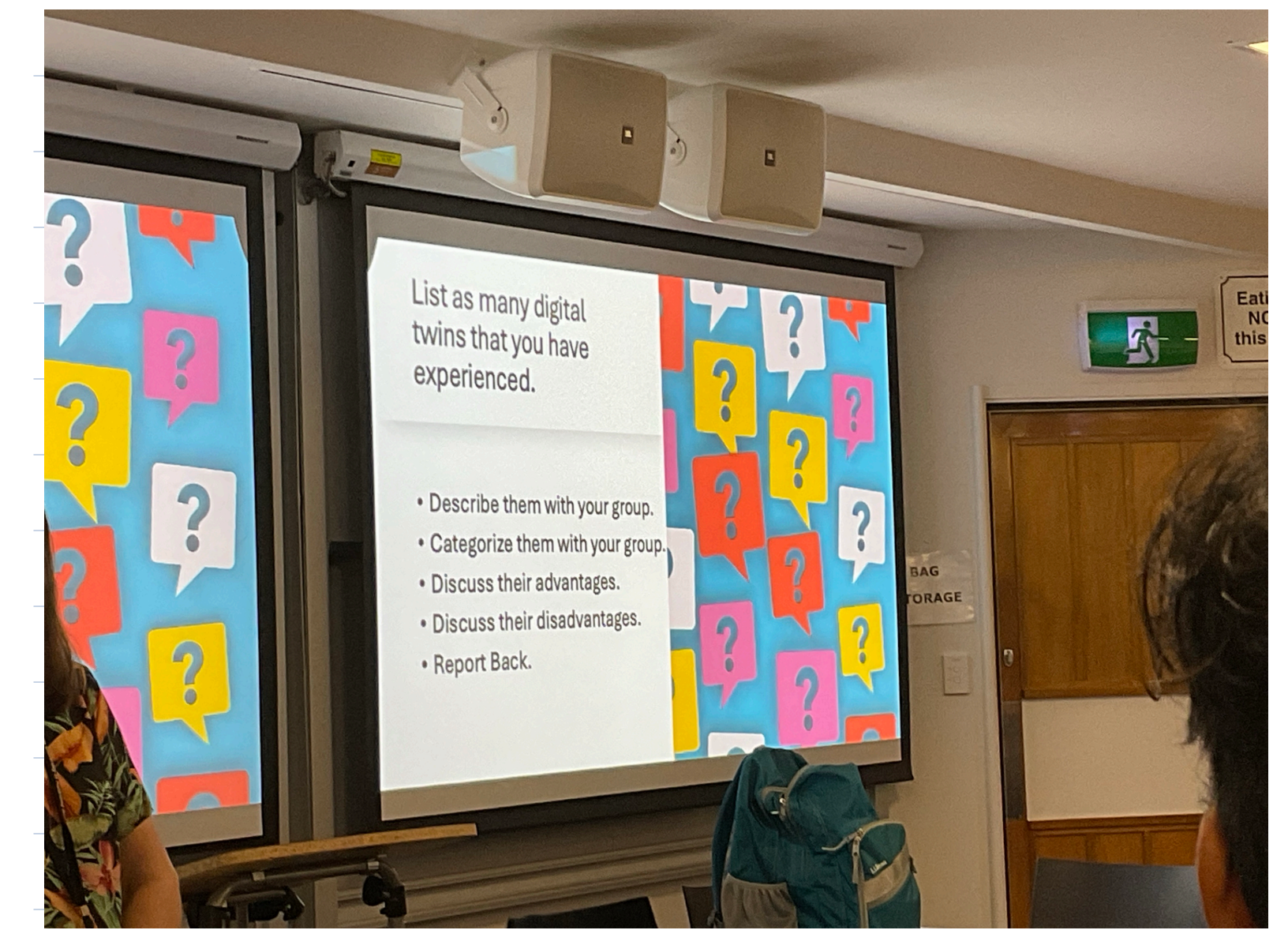
- The bridge between the virtual and the physical world
 - Industry contextual
 - Simulation
 - IOT
 - Data gathering
 - Context
 - Digital representation of physical item
- Information and Cognition
- Dynamically recalibrate the environment design build and operations all affected

The Digital Twin API Development Model





- Characteristics**
- Unique identifier.
 - Sensors and actuators.
 - AI.
 - Communication.
 - Representation.
 - Trust
 - Privacy and security.
- Making DT a reality**
- Augmented, virtual, and mixed reality
 - Haptics
 - Robotics
 - 5G and Tactile Internet
 - Cloud computing
 - Wearables
 - IoT
 - AI



List as many digital twins that you have experienced.

- Describe them with your group.
- Categorize them with your group.
- Discuss their advantages.
- Discuss their disadvantages.
- Report Back.



Figure 1. Virtuality Continuum

Type of Reality	Attributes	Advantages	Disadvantages
Virtual Reality	Full immersion	A sense of presence can be delivered in highly adaptable environments.	Loss of direct interaction with immediate surroundings.
	Not bound by physical laws	Depict what is normally impossible or impractical.	Potentially decreased familiarity of interaction.
Augmented Reality	Mix of virtual and physical objects	Physical world can be enhanced while maintaining sense of presence with immediate surroundings.	Imperfect synching of virtual and physical can decrease effectiveness. Potential information overload.
	Virtual components not bound by physical laws	Virtual components can depict what is normally impossible or impractical.	Potentially decreased familiarity of interaction.
Physical Reality	Sensory richness and seamless immersion	All depictions naturally create a high level of immersion and multiple stimuli provide information on many levels.	Potential of unneeded, distracting stimuli.
	Bound by physical laws	Laws provide a reliable structure for existence.	Laws limit possible activities to those that fall within the bounds of physical law.

Table 1. Summary of attributes for VR, AR, and physical reality